Lab Sheet 1 — DDL Commands: CREATE, ALTER and DROP

1. Create a database named CollegeDB.

```
1  CREATE DATABASE CollegeDB;
2  SHOW DATABASES;
```

Result Grid

Database
CollegeDB
information_schema
mysql
performance_schema
sys

2. Create a Student table with columns: Student_ID, Name, Age, Course, and City.

```
3  USE CollegeDB;
4  CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(100), Age INT,
5    Course VARCHAR(60), City VARCHAR(60));
6  DESCRIBE Student;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
City	varchar(60)	YES		NULL	

3. Create an Employee table with Emp_ID, Emp_Name, Department, and Salary.

```
7  CREATE TABLE Employee (Emp_ID INT PRIMARY KEY, Emp_Name VARCHAR(100), Department VARCHAR(60), Salary
8    DECIMAL(10,2));
9  DESCRIBE Employee;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Emp_ID	int	NO	PRI	NULL	
Emp_Name	varchar(100)	YES		NULL	
Department	varchar(60)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	

4. Create a Department table with Dept_ID, Dept_Name, and Location.

```
10 CREATE TABLE Department (Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(80), Location VARCHAR(80));
11 DESCRIBE Department;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Dept_ID	int	NO	PRI	NULL	
Dept_Name	varchar(80)	YES		NULL	
Location	varchar(80)	YES		NULL	

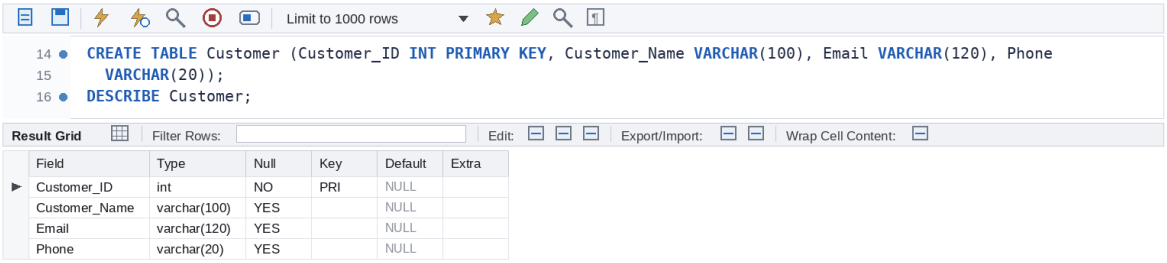
5. Create a Product table with Product_ID, Product_Name, Price, and Quantity.

```
12 CREATE TABLE Product (Product_ID INT PRIMARY KEY, Product_Name VARCHAR(100), Price DECIMAL(10,2), Quantity INT);
13 DESCRIBE Product;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Product_ID	int	NO	PRI	NULL	
Product_Name	varchar(100)	YES		NULL	
Price	decimal(10,2)	YES		NULL	
Quantity	int	YES		NULL	

6. Create a Customer table with Customer_ID, Customer_Name, Email, and Phone.

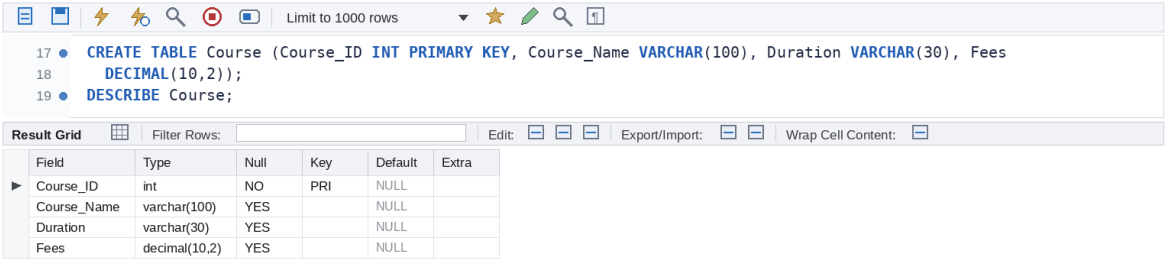


```
14  CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100), Email VARCHAR(120), Phone
15    VARCHAR(20));
16  DESCRIBE Customer;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Customer_ID	int	NO	PRI	NULL	
Customer_Name	varchar(100)	YES		NULL	
Email	varchar(120)	YES		NULL	
Phone	varchar(20)	YES		NULL	

7. Create a Course table with Course_ID, Course_Name, Duration, and Fees.

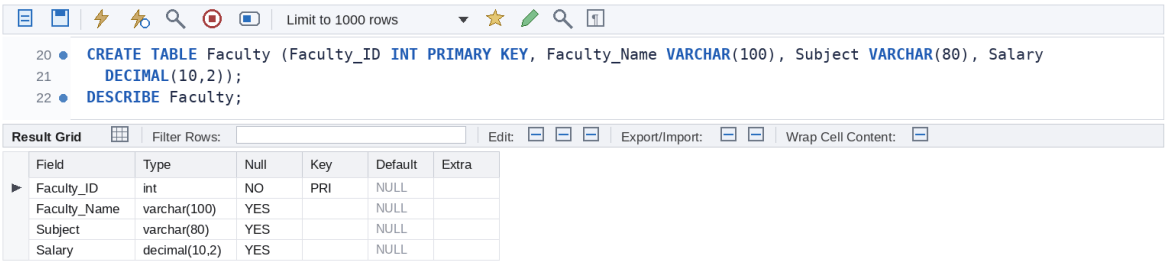


```
17  CREATE TABLE Course (Course_ID INT PRIMARY KEY, Course_Name VARCHAR(100), Duration VARCHAR(30), Fees
18    DECIMAL(10,2));
19  DESCRIBE Course;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Course_ID	int	NO	PRI	NULL	
Course_Name	varchar(100)	YES		NULL	
Duration	varchar(30)	YES		NULL	
Fees	decimal(10,2)	YES		NULL	

8. Create a Faculty table with Faculty_ID, Faculty_Name, Subject, and Salary.

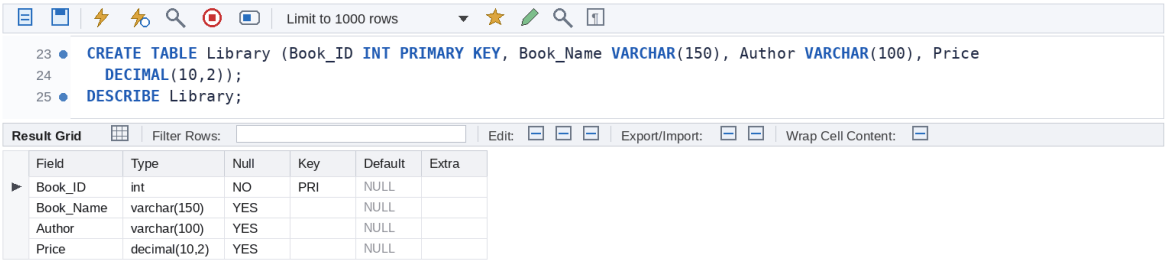


```
20  CREATE TABLE Faculty (Faculty_ID INT PRIMARY KEY, Faculty_Name VARCHAR(100), Subject VARCHAR(80), Salary
21    DECIMAL(10,2));
22  DESCRIBE Faculty;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Faculty_ID	int	NO	PRI	NULL	
Faculty_Name	varchar(100)	YES		NULL	
Subject	varchar(80)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	

9. Create a Library table with Book_ID, Book_Name, Author, and Price.

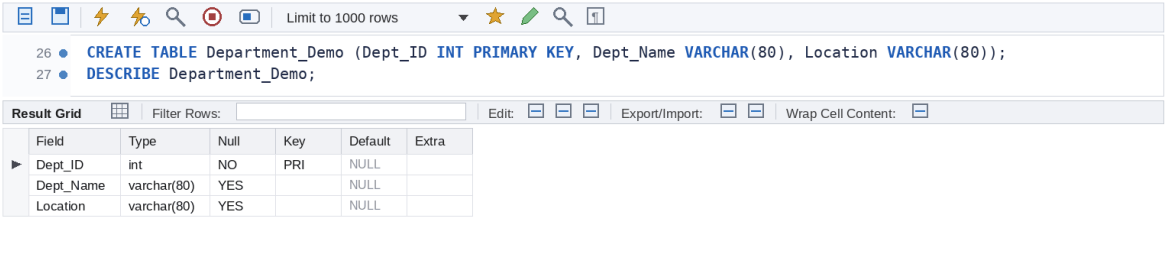


```
23  CREATE TABLE Library (Book_ID INT PRIMARY KEY, Book_Name VARCHAR(150), Author VARCHAR(100), Price
24    DECIMAL(10,2));
25  DESCRIBE Library;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Book_ID	int	NO	PRI	NULL	
Book_Name	varchar(150)	YES		NULL	
Author	varchar(100)	YES		NULL	
Price	decimal(10,2)	YES		NULL	

10. Create a Department table with Dept_ID as the Primary Key.

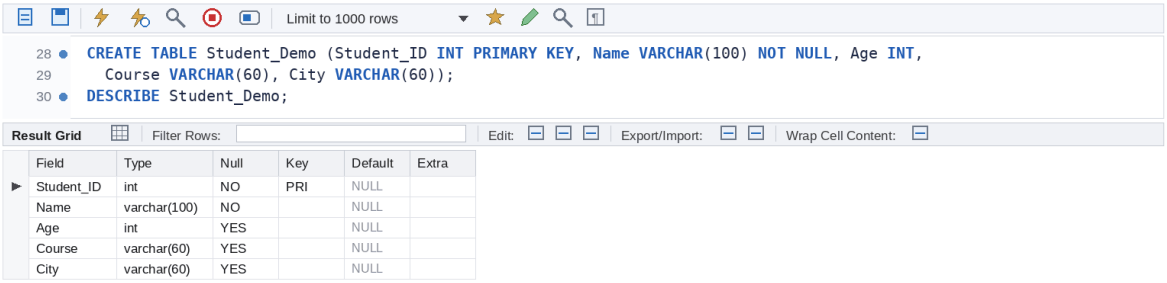


```
26  CREATE TABLE Department_Demo (Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(80), Location VARCHAR(80));
27  DESCRIBE Department_Demo;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Dept_ID	int	NO	PRI	NULL	
Dept_Name	varchar(80)	YES		NULL	
Location	varchar(80)	YES		NULL	

11. Create a Student table with Student_ID as the Primary Key and Name as NOT NULL.

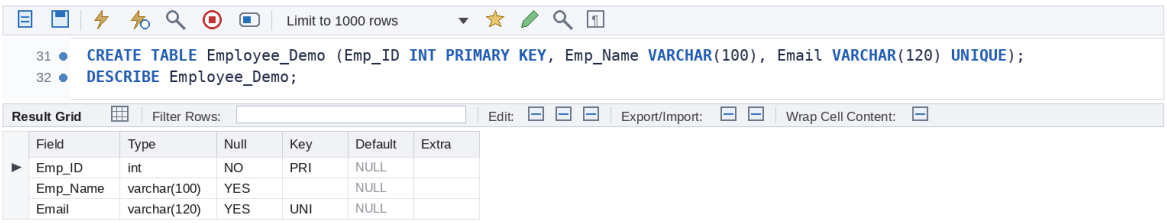


```
28  CREATE TABLE Student_Demo (Student_ID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Age INT,
29    Course VARCHAR(60), City VARCHAR(60));
30  DESCRIBE Student_Demo;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	NO		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
City	varchar(60)	YES		NULL	

12. Create an Employee table with a UNIQUE constraint on Email.

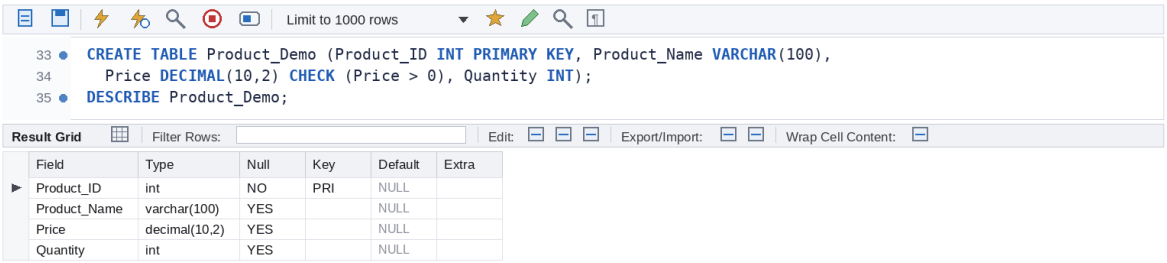


```
31  CREATE TABLE Employee_Demo (Emp_ID INT PRIMARY KEY, Emp_Name VARCHAR(100), Email VARCHAR(120) UNIQUE);
32  DESCRIBE Employee_Demo;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Emp_ID	int	NO	PRI	NULL	
Emp_Name	varchar(100)	YES		NULL	
Email	varchar(120)	YES	UNI	NULL	

13. Create a Product table with a CHECK constraint that Price > 0.

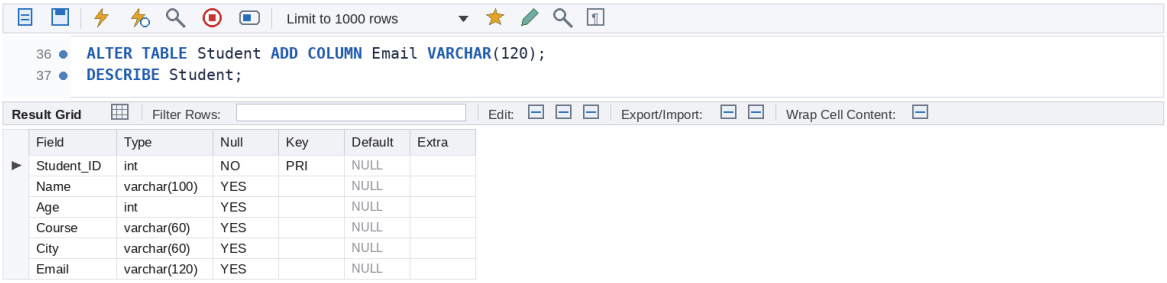


```
33  CREATE TABLE Product_Demo (Product_ID INT PRIMARY KEY, Product_Name VARCHAR(100),
34    Price DECIMAL(10,2) CHECK (Price > 0), Quantity INT);
35  DESCRIBE Product_Demo;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Product_ID	int	NO	PRI	NULL	
Product_Name	varchar(100)	YES		NULL	
Price	decimal(10,2)	YES		NULL	
Quantity	int	YES		NULL	

14. Add a new column Email to the Student table using ALTER TABLE.

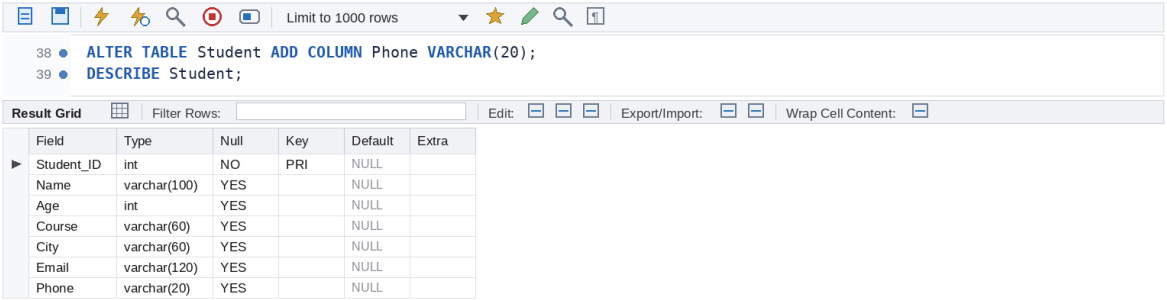


```
36  ALTER TABLE Student ADD COLUMN Email VARCHAR(120);
37  DESCRIBE Student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
City	varchar(60)	YES		NULL	
Email	varchar(120)	YES		NULL	

15. Add a Phone column to the Student table.



```
38  ALTER TABLE Student ADD COLUMN Phone VARCHAR(20);
39  DESCRIBE Student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
City	varchar(60)	YES		NULL	
Email	varchar(120)	YES		NULL	
Phone	varchar(20)	YES		NULL	

16. Modify the datatype/size of the Name column in the Student table.

```
40 ALTER TABLE Student MODIFY COLUMN Name VARCHAR(150);
41 DESCRIBE Student;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(150)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
City	varchar(60)	YES		NULL	
Email	varchar(120)	YES		NULL	
Phone	varchar(20)	YES		NULL	

17. Rename the Student table to Student_Details.

```
42 RENAME TABLE Student TO Student_Details;
43 SHOW TABLES;
```

Result Grid

Tables_in_CollegeDB
Course
Customer
Department
Department_Demo
Employee
Employee_Demo
Faculty
Library
Product
Product_Demo
Student_Demo
Student_Details

18. Rename the column City to Address.

```
44 ALTER TABLE Student_Details RENAME COLUMN City TO Address;
45 DESCRIBE Student_Details;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(150)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
Address	varchar(60)	YES		NULL	
Email	varchar(120)	YES		NULL	
Phone	varchar(20)	YES		NULL	

19. Add a Salary column to the Employee table.

```
46 ALTER TABLE Employee_Demo ADD COLUMN Salary DECIMAL(10,2);
47 DESCRIBE Employee_Demo;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Emp_ID	int	NO	PRI	NULL	
Emp_Name	varchar(100)	YES		NULL	
Email	varchar(120)	YES	UNI	NULL	
Salary	decimal(10,2)	YES		NULL	

20. Drop the Phone column from the Student table.

```
48 ALTER TABLE Student_Details DROP COLUMN Phone;
49 DESCRIBE Student_Details;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(150)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
Address	varchar(60)	YES		NULL	
Email	varchar(120)	YES		NULL	

21. Drop the Email column from the Student table.

```
50  ALTER TABLE Student_Details DROP COLUMN Email;
51  DESCRIBE Student_Details;
```

Result Grid

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(150)	YES		NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
Address	varchar(60)	YES		NULL	

22. Remove all records from a table using TRUNCATE.

```
52  INSERT INTO Student_Details VALUES (1, 'Anshika', 20, 'B.Tech CSE', 'Jaipur'),
53    (2, 'Radha', 19, 'B.Tech CSE', 'Kashipur');
54  TRUNCATE TABLE Student_Details;
55  SELECT * FROM Student_Details;
```

Result Grid

Student_ID	Name	Age	Course	Address
NULL	NULL	NULL	NULL	NULL

23. Drop the Course table from the database.

```
56  DROP TABLE Course;
57  SHOW TABLES;
```

Result Grid

Tables_in_CollegeDB
Customer
Department
Department_Demo
Employee
Employee_Demo
Faculty
Library
Product
Product_Demo
Student_Demo
Student_Details

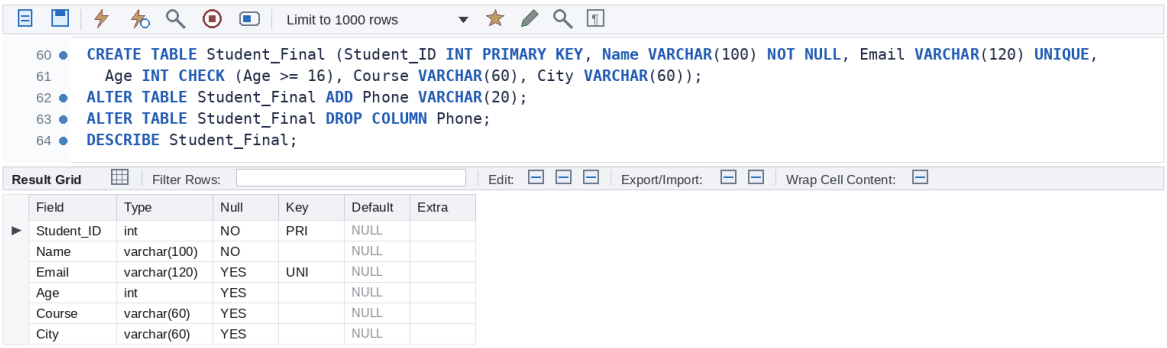
24. Drop the Customer table from the database.

```
58  DROP TABLE Customer;
59  SHOW TABLES;
```

Result Grid

Tables_in_CollegeDB
Department
Department_Demo
Employee
Employee_Demo
Faculty
Library
Product
Product_Demo
Student_Demo
Student_Details

25. Create a complete Student table using Primary Key, NOT NULL, UNIQUE and CHECK constraints.



```
60  CREATE TABLE Student_Final (Student_ID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Email VARCHAR(120) UNIQUE,
61    Age INT CHECK (Age >= 16), Course VARCHAR(60), City VARCHAR(60));
62  ALTER TABLE Student_Final ADD Phone VARCHAR(20);
63  ALTER TABLE Student_Final DROP COLUMN Phone;
64  DESCRIBE Student_Final;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Student_ID	int	NO	PRI	NULL	
Name	varchar(100)	NO		NULL	
Email	varchar(120)	YES	UNI	NULL	
Age	int	YES		NULL	
Course	varchar(60)	YES		NULL	
City	varchar(60)	YES		NULL	